

STATISTICAL ANALYSIS PLAN

SAP — Cognitive Behavioral Therapy (CBT) in Adults with moderate-to-severe depression (HAM-D ≥ 17)

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Research question

/sap two-arm RCT of CBT vs SSRI for depression, primary outcome HAM-D at 12 weeks

Trial design (PICOT)

Population	Adults with moderate-to-severe depression (HAM-D ≥ 17)
Intervention	Cognitive Behavioral Therapy (CBT)
Comparator	SSRI (selective serotonin reuptake inhibitor)
Primary outcome	Hamilton Depression Rating Scale (HAM-D) score change from baseline to 12 weeks
Secondary outcomes	—
Timeframe	Primary outcome assessment at 12 weeks; total follow-up 12 weeks
Design	parallel rct
Hypothesis	superiority
Outcome type	continuous

Sample-size derivation

Formula	Two-sample t-test (Cohen, common SD)
n per arm (control)	77
n per arm (intervention)	77
Total enrolment	154
Reference	Cohen J. Statistical Power Analysis for the Behavioral Sciences, 2nd ed. Erlbaum, 1988. §2.4.

Analysis plan (ICH E9)

Populations: ITT / Safety

Primary test: Two-sample t-test (ANCOVA with baseline HAM-D covariate)

Comparison of HAM-D score change from baseline to 12 weeks between CBT and SSRI arms using a two-sample t-test, with baseline HAM-D score as a covariate. Analysis conducted on the intention-to-treat population.

Multiplicity strategy: none

Missing data: multiple imputation mar

Sensitivity analyses

- Per-protocol analysis excluding significant protocol violators
- On-treatment analysis censoring follow-up after treatment discontinuation
- Tipping-point sensitivity analysis for missing data under MNAR assumptions

Subgroup analyses

- Age group (≤ 40 vs > 40 years)
- Baseline HAM-D severity (17–23 vs ≥ 24)
- Comorbid anxiety disorder (yes vs no)

Safety monitoring

Safety population (all randomized participants who receive at least one dose) will be monitored for adverse events (AE) and serious adverse events (SAE). A Data Monitoring Committee will review SAE reports monthly. Stopping rules for safety: trial halted if ≥ 3 unexpected serious adverse reactions attributable to study treatment are reported within any 4-week period.

Background

Major depressive disorder (MDD) remains a leading cause of disability worldwide, with substantial heterogeneity in treatment response. Both cognitive behavioral therapy (CBT) and selective serotonin reuptake inhibitors (SSRIs) are established first-line treatments; however, head-to-head comparison of their efficacy on symptom severity change is limited. This parallel-group, two-arm randomized controlled trial aims to test the hypothesis that CBT is superior to SSRI monotherapy in reducing depressive symptoms (measured by the Hamilton Depression Rating Scale) at 12 weeks post-randomization in adults with moderate-to-severe major depressive disorder. The Hamilton Depression Rating Scale (HAM-D) is a well-validated clinician-administered outcome measure with established reliability and sensitivity to change in depression severity.

References

- [1] Statistical Power Analysis for the Behavioral Sciences *Cohen J, 1988*